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characters	words	sentences	reading time	speaking time

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Issues left

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22 **Clarity**

21 Passive voice misuse

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The Ethical Issue of Facial Recognition and Privacy

Isabella Herrera Pilonieta

Department of Science & Mathematics: Huntington University

CSC362: AI and Machine Learning

Dr. Jeff Lehman

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Abstract

Facial recognition technology has raised significant ethical questions about security and privacy. The Integrated Joint Operation Platform in China is a large-scale surveillance program that is the subject of this paper's analysis of the ethical implications of facial recognition technology. IT investigates how these technologies may violate people's rights by enabling continuous monitoring and automated decision-making, as well as by collecting and analyzing biometric data without consent. The essay uses legal frameworks, including the Personal Information Protection Law, and finds differences between the law and its actual implementation. Recent efforts to address these issues are also discussed[1] in this essay, including the use of new facial recognition methods that ensure user consent, data security, and transparency. The paper includes a biblical perspective on human dignity, justice, and fairness as a foundation for evaluating the ethical use of technology. Finally, while facial Recognition has significant potential to reduce crime, AI must be used carefully, aligned with ethical standards, and respectful of human rights.

The Ethical Issue of Facial Recognition and Privacy

Facial Recognition raises significant concerns in the context of AI and Machine Learning (ML). To recognize individual faces based on patterns and facial traits, this technology uses deep learning algorithms that analyze and compare a vast library of photos. This issue is important because governments and private companies are adopting facial recognition systems to replace ID cards or fingerprints, and often without individuals' knowledge or consent. For example, some "property management companies installing facial recognition software at the entrances to apartment complexes without the residents' permission, heightening citizens' awareness of the dangers of losing their face." (China's Facial Recognition Concerns, s. f.). While many individuals voluntarily use facial Recognition and similar technologies on their personal devices, non-consensual use raises ethical concerns. These concerns include questions like,

"Where is this data going?" Is it secure enough? How is data management working when companies use people's faces without their consent? Is it legal? As this technology grows, we see the importance of addressing privacy rights and protecting individuals.

Facial Recognition Technology works.

Facial Recognition uses machine learning and neural networks to detect faces by extracting patterns and measurements of people's faces. The algorithm I will use is the Dlib CNN detector, which is a deep learning-based Maximum-Margin Object Detection (MMOD) algorithm. How the algorithm works is the following (Face Recognition Using Artificial Intelligence):

Face detection: Identify the region where a face is present.

Face alignment: Uses key facial landmarks such as nose, eyes, and lips.

Feature extraction: The deep learning model converts the landmarks into numerical vectors, called embeddings, and compares the embeddings of two faces to determine their similarity.

Face matching: the system compares them to a different person by using different techniques.

Facial Recognition and its evolution

Facial Recognition has been explored[2] since the early 1960s. Woody Bledsoe developed a semi-automated system that required humans to use their facial features, like eyes, ears, and nose, to match them to stored images. In the 1980s, Sirovich and Kirby developed the Eigenfaces, which reduced the complexity of facial images and distinguished one face from another. In the early 2000s, the U.S. Department of Defense developed a standardized method for evaluating facial requirements. In 2010, the development of Convolutional Neural Networks (CNNs) transformed the field, enabling it to process large

amounts of visual data. In 2014, DeepFace, developed by Facebook, set a benchmark of 97% accuracy in facial recognition tasks.

Neural Networks

Deep learning techniques use neural networks, which are a subset of machine learning. They are composed of node levels, each with an input layer, an output layer, and one or more hidden layers. Each node has a connected threshold and a weight. If a node's output exceeds the specified threshold, it becomes active and sends data to the subsequent layer of the network. If not, no data is moved[3] to the next tier of the network. Prior to CNNs, tedious, manual feature extraction methods were used to identify objects in photos. These days, CNNs provide a more scalable approach to object identification and image classification by using linear algebraic operations, particularly matrix multiplication, to find patterns in images. However, they can be computationally demanding because they are trained on graphical processing units (GPUs) (What Are Convolutional Neural Networks?, 2021).

Case Analysis

One example of the increase in AI and facial Recognition is seen[4] in China. A student in Beijing reported that the school had installed a facial recognition camera in the dormitories

without any sense of permission or consent from the students. At the same time, a

professor at Zhou's university reported that the university added cameras in the teaching

buildings without his consent to share his facial data (China's Facial Recognition Concerns, s. f.). This ethical issue is growing because there are no regulations governing the use of biometric data. It is unclear who is responsible for keeping biometric data safe, allowing it to be leaked and used in scams and

to steal identities if not managed properly. This problem keeps growing, reaching a point where there is no freedom or privacy, whether you are on the street or in your own apartment. There are about 350 million surveillance cameras in China as of 2018; this number is forecast[5] to keep growing as new and more advanced facial recognition technology is implemented[6].

In China, there is an algorithm referred to as "China's Algorithm of Repression". In a Human Rights Watch report, we can see that in Xinjiang, China, the government built an AI-powered mass surveillance system. This app collects information from individuals connected to IJOP (China's system). This system uses facial Recognition and Wi-Fi sensors to collect data. Some of the information this application collects includes name, ID number, phone usage, car details, electricity usage, religion, and social relationships. The AI's purpose is to flag in the system behaviors that are not usual; some examples Human Rights Watch emphasizes include using VPNs, WhatsApp, traveling outside the area, and having relatives abroad. All these behaviors are legal, but the AI might flag them as "suspicious." Whenever there is suspicious activity, the app sends an alert to the police through the app, and the police can decide to investigate. Some of these cases end up in arrests. Another tool this AI has is facial detection to check whether a photo matches an ID through cross-referencing, including recognizing people from both national and foreign embassies. People are being recorded[7] and analyzed every minute of their daily lives; if they behave outside their "routine," the system could flag it as "suspicious."

Negative instances of Facial Recognition

China's app raises many ethical concerns, particularly the IJOP; for instance, a main concern is the lack of consent to share personal and biometric information. Most people are unaware that their personal data is being collected and do not have the opportunity to agree or deny. According to the

Personal Information Protection Law, Article 29 says, "Individual consent should be obtained for processing sensitive personal information. Where laws and administrative regulations provide that the processing of sensitive personal information shall be subject to written consent, such provisions shall prevail" (Law, 2022b). The law seeks to protect people's biometric information. However, extensive monitoring is conducted without each person's express consent, raising concerns about whether such practices comply with the law, particularly when applied to an entire population in public areas.

The application also violates the principle of "innocent until proven guilty," and basic human activities like traveling outside the country or changing your routine can be flagged[8] as suspicious rather than reflecting growth or change as a person. These behaviors are normal and do not mean wrongdoing. However, the system can interpret them as potential threats, leading to investigations or even arrest. This approach changes the situation for individuals trying to prove their innocence when conflicting fundamental ethical and legal principles are at stake.

Additionally, the IJIOP system automatically determines whether the behavior is normal or abnormal. Even if it is harmless and there is no crime, the AI could still flag it. AI systems do not fully understand human interacts and complexity; the lack of context or intent plays an important role in why humans act the way they do. As a result, harmless actions can be interpreted as wrongdoings and viewed as suspicious, leading to severe consequences or investigations. Article 24 of the PIPL states that individuals have the right to understand the decisions made by automated processes. However, the IJOP system offers little transparency or the opportunity for individuals to contest these decisions.

Finally, the system also collects private information, such as relationships, personal habits, and daily activities, which has raised privacy concerns. Article

28 of the PIPL states that "personal information processors can process sensitive personal information only when they have a specific purpose and sufficient necessity, and take strict protective measures" (Law, 2022a). However, the extent and depth of data collection in these public places raise questions about their necessity and proportionality. Monitoring individuals' movements and behaviors daily can be seen as invasive surveillance rather than a security effort, challenging the rationale for the government's use of this technology.

Positive efforts to mitigate Facial Recognition concerns

China has also sought to implement new rules and regulations governing facial Recognition. To safeguard personal information and prevent misused, the Security Management Measures for the Application of Facial Recognition Technology were established[9] in 2025. Starting with storage, it prohibits external transfers unless the data subject gives separate authorization and relevant laws and regulations permit the transfer. Before processing the data, the Privacy Impact Assessment ("PIA") must also contact the subject.

The measure also adds a limitation on public places where facial recognition devices can be installed[10]; however, it must be subject to the data handler establishing the necessity for maintaining public security. The data handler shall reasonably determine the area for facial information collection and display prominent warning signs(China Releases New Rules Regarding the Use of Facial Recognition Technology, s. f.). In addition, the data handler should not rely solely on facial Recognition as the verification method if another technology can achieve the same purpose or meet equivalent business requirements. Companies with more than 100,000 individuals that use facial recognition technology should file with the competent Cyberspace authority at the provincial level or higher within 30 business days upon reaching that

threshold (China Releases New Rules Regarding the Use of Facial Recognition Technology, s. f.).

In addition to these regulations, China aims to address concerns about data security and unauthorized surveillance. To protect people's privacy, businesses should avoid using facial Recognition in sensitive areas such as hotels, public restrooms, and private places. They should also explain the reasons and purposes for using biometric data. Additionally, companies must be open about how they use and collect biometric data, including how long it takes and how it can affect specific people. These notices must be simple to understand, and the company is obligated[11] by law to notify people if the data it uses changes. Prior to collection, specific parental consent is required[12] for special protections for kids' facial data. Companies must implement additional security measures and follow specific rules on data use, retention, and transfer to protect their data and prevent misuse.

Finally, the measures introduce more security and risk management; there are strict regulations on data encryption, access controls, regular security audits to identify vulnerabilities, risk assessments, and incident response protocols to implement corrective actions.

Facial Recognition VS Faith

Genesis 1:27 emphasizes that as humans, "God created man in his own image, in the image of God he created him; male and female he created them" (Genesis 1:27 NIV - So God created mankind[13] in his own - Bible Gateway, s. f.). This idea shows that every human should be treated[14] with respect, autonomy, and morality. When we talk about Facial Recognition, this raises concerns about the systems monitoring individuals without consent or reducing people to data points. Continuous surveillance can undermine the sense of freedom and personal dignity that come from being recognized as individuals created with

purpose and value. From this perspective, the use of AI needs to be carefully considered[15] to ensure it does not dehumanize or exploit individuals. Additionally, the Bible highlights the importance of fairness and justice. Proverbs 21:3 states, "To do righteousness and justice is more acceptable to the LORD than sacrifice" (Proverbs 21:3 NIV - To do what is right and just is more - Bible Gateway, s. f.). These principles challenge the use of facial Recognition because the system may unfairly target or flag individuals due to bias. Systems with large surveillance platforms risk treating people suspiciously without evidence, which conflicts with biblical principles. The ethical use of technology should be prioritized, with individuals protected, rather than creating and reinforcing systems that may lead to unjust outcomes.

Personal Faith Reflection

One of my closest experiences with biometric data occurred at the airport in Colombia, where you can skip traditional immigration by registering in a biometric system that scans your eyes. This system allows travelers to pass through security without standing in long lines and interacting with an officer. My second experience was in college, when I applied for OPT through U.S. Citizenship and Immigration Services (USCIS). As part of the process, I was required[16] to provide a photo as biometric data. In both cases, the biometric data was reasonable, limited to a specific purpose, and did not include day or night surveillance.

From my personal faith, human dignity and freedom are essential values that guide technology. While I understand that governments are responsible for public safety, there needs to be clear limits on the use of facial Recognition. One example will be using biometric systems to investigate cases of crime or other violations. However, recording and recognizing people in their daily lives is

a complete violation of privacy; freedom will no longer be freedom, and the government will manage people's lives.

At the same time, facial Recognition is going to keep growing and will be used[17] more often than it is now. When applied properly, AI technology can deliver great results. It might contribute to a decrease in crimes, including human trafficking, violence, and kidnappings. From a faith perspective, technology should serve humanity, not control it; instead of monitoring ordinary people, companies should direct AI to address real issues and protect vulnerable people. In this way, technology will be used[18] ethically and can align power with the values of justice, compassion, and respect for human life.

Conclusion and Solutions

To conclude, Facial recognition technology is a powerful and sensitive topic that can get out of hand if not used properly. The essay discusses the advantages and disadvantages of facial recognition technology. For example, when used appropriately, it could reduce crime such as violence, theft, and human trafficking. However, systems like the IJOP demonstrate how technology can be abused[19] to monitor people, restricting their freedom and leading to unfair choices motivated by prejudice. In the absence of appropriate regulations, facial recognition technology may become a control mechanism rather than a tool for protection, posing grave ethical questions about privacy, human rights, and fairness.

To address some concerns, first, biometric data should be stored in a highly secure location, using encryption and strict access controls to prevent data breaches. Government and organizations should ensure data collection is limited to what is necessary and obtain consent from the subjects. Additionally, protections should be put[20] in place for minors, including parental consent and enhanced security for their biometric data. In public spaces, people need to be

aware that they are being filmed, like in stores or malls. This[21] will allow people to trust that, in an emergency, they can recall the events.

From a faith perspective, these solutions are aligned[22] with biblical principles that ensure justice and fairness. Technology should be used[23] to serve and protect people, not control us. Respecting privacy reflects human dignity while ensuring fairness in AI systems. By always putting people first and ensuring everyone is safe, society can achieve a balance in which facial Recognition contributes to safety and progress without compromising fundamental human rights and freedoms.

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1.	*are also discussed*	Passive voice misuse	Clarity
2.	*been explored*	Passive voice misuse	Clarity
3.	*is moved*	Passive voice misuse	Clarity
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7.	*being recorded*	Passive voice misuse	Clarity
8.	*basic human activities like traveling outside the country or changing your routine can be flagged*	Passive voice misuse	Clarity
9.	*the Security Management Measures for the Application of Facial Recognition Technology were established*	Passive voice misuse	Clarity
10.	*facial recognition devices can be installed*	Passive voice misuse	Clarity
11.	*is obligated*	Passive voice misuse	Clarity
12.	*is required*	Passive voice misuse	Clarity
13.	~~mankind~~ → humanity, humankind	Potentially sensitive language	Delivery
14.	*be treated*	Passive voice misuse	Clarity
15.	*be carefully considered*	Passive voice misuse	Clarity
16.	*was required*	Passive voice misuse	Clarity
17.	*be used*	Passive voice misuse	Clarity
18.	*technology will be used*	Passive voice misuse	Clarity

19.	*be abused*	Passive voice misuse	Clarity
20.	*protections should be put*	Passive voice misuse	Clarity
21.	*This*	Intricate text	Clarity
22.	*are aligned*	Passive voice misuse	Clarity
23.	*Technology should be used*	Passive voice misuse	Clarity
24.	~~mankind~~ → humanity, humankind	Potentially sensitive language	Delivery
25.	*2 The Ethical Issue of Facial Recognition and Privacy Isabella Herrera Pilonieta Department of Science & Mathematics: Huntington University CSC362: AI and Machine Learning Dr. Jeff Lehman April 2, 2026 Abstract Facial recognition technology has raised significant ethical ques...*		Originality
26.	*l Issue of Facial Recognition and Privacy Facial Recognition raises significant concerns in the context of AI and Machine Learning (ML). To recognize individual faces based on patterns and facial traits, this technology uses deep learning algorithms that anal*		Originality
27.	*with an input layer, an output layer, and one or more hidden layers.*	TensorFlow tutorial: Get started with TensorFlow machine learning | InfoWorld https://www.infoworld.com/article/3138908/tensorflow-tutorial-get-started-with-tensorflow.html	Originality
28.	*Individual consent should be obtained for processing sensitive personal information. Where laws and administrative regulations provide that the processing of sensitive personal information shall be subject to written consent, such provisions shall prevail*	Rules for Processing Sensitive Personal Information Archives - PIPL https://personalinformationprotectionlaw.com/PIPL/category/rules-for-processing-sensitive-personal-information/	Originality

29.	*s to protect people's biometric information. However, extensive monitoring is conducted without each person's express consent, raising concerns about whether such practices comply with the law, particularly when appl*		Originality
30.	*ied to an entire population in public areas. The application also violates the principle of "innocent until proven guilty," and basic human activities like traveling outside the country or changing your routine can be flagged as suspicious rather than reflecting growth or change as a person. These ...*		Originality
31.	*personal information processors can process sensitive personal information only when they have a specific purpose and sufficient necessity, and take strict protective measures*	Rules for Processing Sensitive Personal Information Archives - PIPL https://personalinformationprotectionlaw.com/PIPL/category/rules-for-processing-sensitive-personal-information/	Originality
32.	*the Security Management Measures for the Application of Facial Recognition Technology*	China's Facial Recognition Regulations Set to Take Effect June 1 https://natlawreview.com/article/china-releases-new-rules-regarding-use-facial-recognition-technology	Originality
33.	*with the competent Cyberspace authority at the provincial level or higher within 30 business days upon reaching that threshold*	China's Facial Recognition Regulations Set to Take Effect June 1 https://natlawreview.com/article/china-releases-new-rules-regarding-use-facial-recognition-technology	Originality

34.	*China Releases New Rules Regarding the Use of Facial Recognition Technology,*	China's Facial Recognition Regulations Set to Take Effect June 1 https://natlawreview.com/article/china-releases-new-rules-regarding-use-facial-recognition-technology	Originality
35.	*s. f.). In addition to these regulations, China aims to address concerns about data security and unauthorized surveillance. To protect people's privacy, businesses should avoid using*		Originality
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38.	*God created man in his own image, in the image of God he created him; male and female he created them" (Genesis 1:27*	The Triumvirate Of Human Development: A Balanced Approach To Life	Originality
39.	*NIV - So God created mankind in his own - Bible Gateway,*	Feminist Readings Of Evangelicalism: Views On Gender And Sexuality From Evangelical Perspectives	Originality
40.	*the use of AI needs to be carefully considered to ensure it does not*	Assessing AI Integration in Islamic Higher Education: A Mixed-Methods Fishbone Diagram Analysis	Originality

41.	*To do righteousness and justice is more acceptable to the LORD than sacrifice" (Proverbs 21:3*	Proverbs 21:3 - Bible verse - DailyVerses.net https://dailyverses.net/proverbs/21/3	Originality
42.	*d disadvantages of facial recognition technology. For example, when used appropriately, it could reduce crime such as violence, theft, and human trafficking. However, systems like the IJOP demonstrate how technology can be abused to monitor people, restricting their freedom and*		Originality
43.	*encryption and strict access controls to prevent data*	Strengthening the Principle of Beneficence in Safeguarding Patient Data Confidentiality: An Analysis of the Roles and Responsibilities of Hospitals	Originality
44.	*China Releases New Rules Regarding the Use of Facial Recognition Technology.*	China's Facial Recognition Regulations Set to Take Effect June 1 https://natlawreview.com/article/china-releases-new-rules-regarding-use-facial-recognition-technology	Originality
45.	*Genesis 1:27 NIV - So God created mankind in his own*	Feminist Readings Of Evangelicalism: Views On Gender And Sexuality From Evangelical Perspectives	Originality